CPU

- Hardware is the physical component of a computer.
- Tech continues to evolve rapidly by:
  - Increased capacity, performance and reduced cost.

Year	Technology	Relative performance/cost
1951	Vacuum tube	1
1965	Transistor	35
1975	Integrated circuit (IC)	900
1995	Very large scale IC (VLSI)	2,400,000
2013	Ultra large scale IC	250,000,000,000

Figure 1: Key Takeaway: performance has increased exponentially while cost has decreased

Inside the PC

- Computer systems have four parts:
  1. Hardware: physical components (e.g., CPU, RAM, Disk, Mouse, Keyboard)
  2. Software: system software, applications, middleware
     - Middleware: Software that acts as an intermediary between different systems or applications, facilitating their interaction and communication.
  3. Information processed by the system
  4. User: people involved in using the system

Inside the CPU

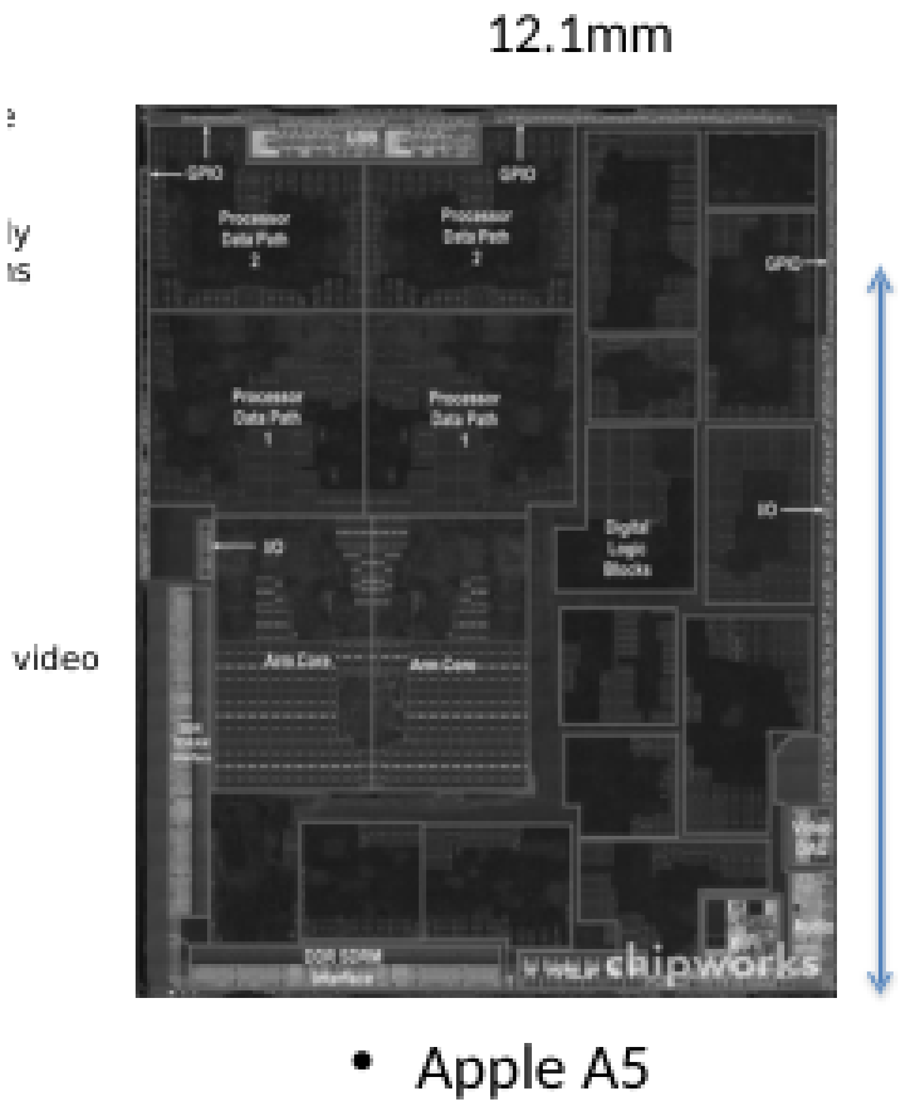

Figure 2: Apple A5 CPU

- The CPU has many core configurations. Here are some:
- Single core: One processing unit.
- Dual core: Two processing units.
- Quad core: Four processing units.
- Octa core: Eight processing units.
- Multi-core: General term for processors containing two or more independent cores.

Furthermore, there is deca-core (for 10 core)…, basically the more the cores the better the multitasking and parallel processing.

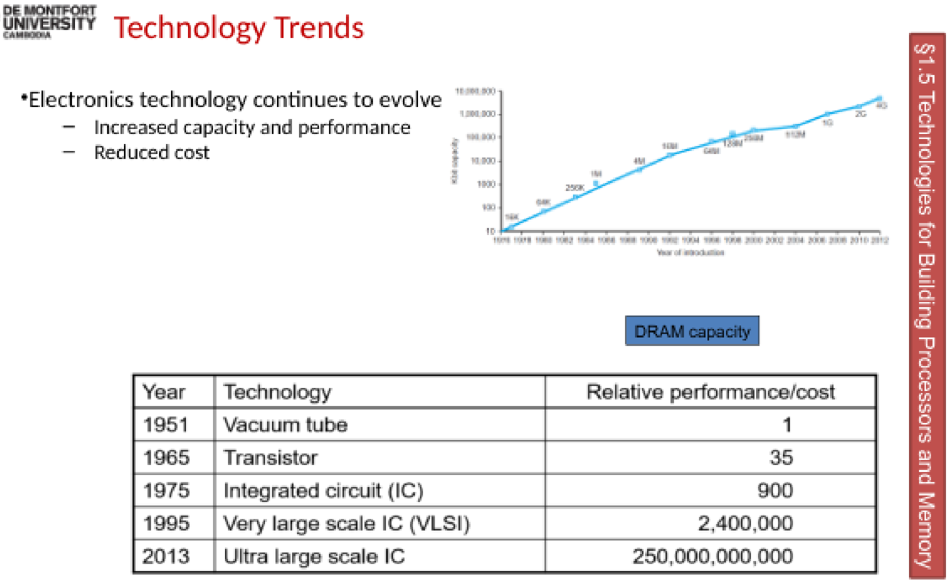

Figure 3: Tech trends in recent decades

Moore's Law

Gordon Moore (co-founder of Intel) observed in 1965 that the number of transistors on a microchip doubles approximately every two years, while the cost of computers is halved. This trend has driven the rapid advancement of computing power for decades, though it is currently slowing down as we approach the physical limits of silicon manufacturing.

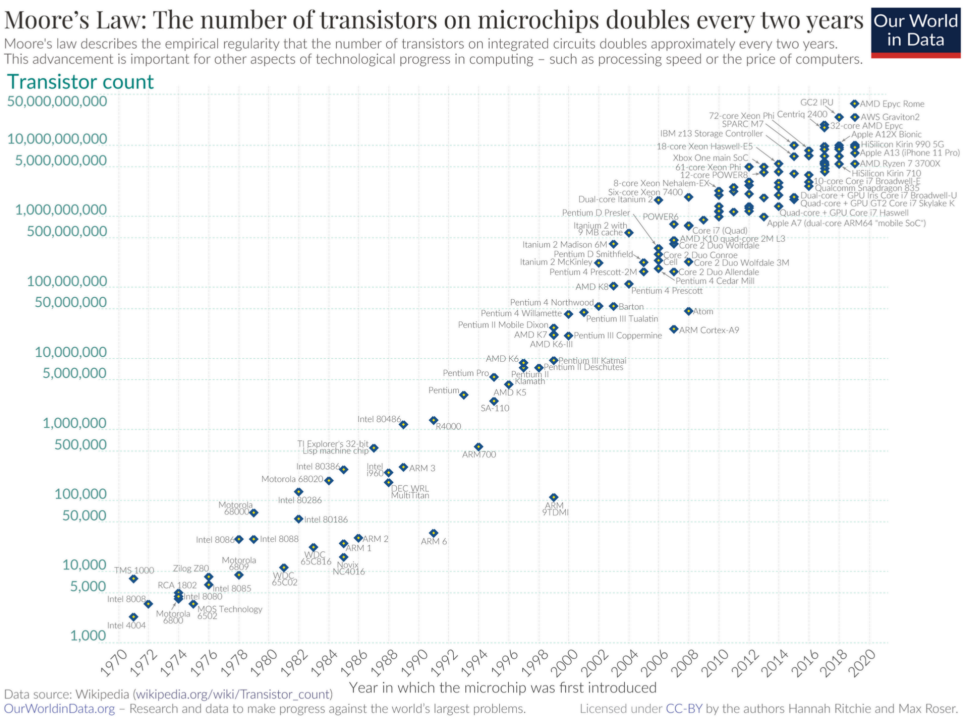

Figure 4: As of 2015: progress has slowed, doubling every 2.5 years

Logic Gates

Computer are built of tiny switches (that can turn on (1) and off (0)) called transistors, by combining a bunch of transistors you can get **logic gates**.

Examples:

- AND : Output is 1 only if both inputs are 1.
- OR : Output is 1 if either both inputs are 1.
- NOT : Flips the input ($1 \to 0, 0 \to 1$)
- NAND & NOR : All in one circuits, meaning they are functionally complete.

Combination Circuits are circuits that only look at the current input (Ex: Logic Gates). They don't "remember" anything.

Sequential Circuits, on the other hand, looks at the current input and what happened in the past. To do this, they use Flip-Flops, which act as a 1 bit cell.

To watch: https://www.youtube.com/watch?v=Hi7rK0hZnfc

Memory

- SRAM (Static RAM): Used for Cache (close to the CPU). It is very fast, expensive, and uses multiple transistors per bit, not just one. It does not need to be refreshed.
- DRAM (Dynamic RAM): Used in Main Memory. It is slower but cheaper, denser, and users a capacitor to store charges. Since these capacitor can leak, they need to be refreshed.
- Flash & Disk: Used for long term storage.

Practice

1. Given the MIPS instruction give a run down on the 5 CPU stages to make this instruction happen.

```
add $s0, $s1, $s2
```

Stage	Action for `add $s0, $s1, $s2`
IF	Fetch the instruction from memory and increment the Program Counter.
ID	Decode instruction; read values of `$s1` and `$s2` from the Register File.
EX	ALU calculates the sum of the values from `$s1` and `$s2`.
MEM	No memory access needed; the sum simply passes through this stage.
WB	Write the sum result back into the register `$s0`.

2. Convert the C code into MIPS ASM.

```
if ( i != j ) f = g + h;
else f = g - h;
```

Solution:

```
# Compare i and j
    bne $s3, $s4, L1    # if (i != j) goto L1

    # Else block: f = g - h
    sub $s0, $s1, $s2   # f = g - h
    j L2                # jump to end (skip L1)

L1: # If block: f = g + h
    add $s0, $s1, $s2   # f = g + h

L2: # End
```

3. Fill in the cells

	1. Arithmetic	2. Load	3. Store	4. Branch
Registers	✓	✓	✓	✓
ALU	✓	✓	✓	✓
Data Memory	x	✓	✓	x
Control Unit	✓	✓	✓	✓
PC	✓	✓	✓	✓